

All about 'this'

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1. The 'this' Keyword?

- The 'this' pointer(C++)/reference(Java) is a special pointer available inside **non-static** member functions of a **class**.
- It points to the **object** that is currently calling the member function.

```
#include <iostream>
using namespace std;

class Person {
    string name;
public:
    Person(string n) {
        name = n;
    }

    void showName() {
        cout << "My name is " << this->name << "\n";
    }
};
```

- **this->name** means "the name which belongs to **current object**".
- If we do...

```
Person p1("Alice");
p1.showName(); // prints: My name is Alice

Person p2("Bob");
p2.showName(); // prints: My name is Bob
```

- Prints...

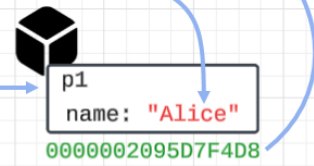
```
/*
Address of p1: 0000002095D7F4D8
Address of this: 0000002095D7F4D8
My name is Alice
-----
Address of p2: 0000002095D7F518
Address of this: 0000002095D7F518
My name is Bob
*/
```

```

class Person {
    string name;
public:
    Person(string n) {
        name = n;
    }
    void showName() {
        cout << "Address of this: " << this << "\n";
        cout << "My name is " << this->name << "\n";
    }
};

int main() {
    Person p1("Alice");
    cout << "Address of p1: " << &p1 << "\n";
    p1.showName();
    cout << "-----\n";
    Person p2("Bob");
    cout << "Address of p2: " << &p2 << "\n";
    p2.showName();
}

```

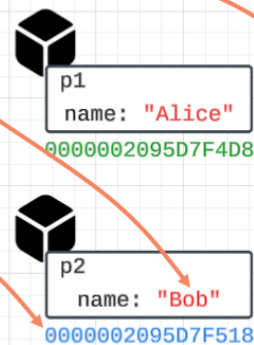


```

class Person {
    string name;
public:
    Person(string n) {
        name = n;
    }
    void showName() {
        cout << "Address of this: " << this << "\n";
        cout << "My name is " << this->name << "\n";
    }
};

int main() {
    Person p1("Alice");
    cout << "Address of p1: " << &p1 << "\n";
    p1.showName();
    cout << "-----\n";
    Person p2("Bob");
    cout << "Address of p2: " << &p2 << "\n";
    p2.showName();
}

```



Why Use `this`?

- To refer to the current object, especially when:
 - You have naming conflicts between parameters and member variables.
 - You need to return the current object from a function.

Naming conflicts:

- When the **parameter name** is the **same as a member variable**

```
class MyClass {
    int a;
public:
    void setValue(int a) {
        a = a;
    }
    void showValue() {
        cout << "Value of a: " << a;
    }
};

int main() {
    MyClass obj;
    obj.setValue(10);
    obj.showValue();
}
// Value of a: -858993460
```

```
class MyClass {
    int a;
public:
    void setValue(int a) {
        this->a = a;
    }
    void showValue() {
        cout << "Value of a: " << this->a;
    }
};

int main() {
    MyClass obj;
    obj.setValue(10);
    obj.showValue();
}
// Value of a: 10
```

```
class Student {
    String name;

    Student(String name) {
        name = name;
    }

    void show() {
        System.out.println("Name: " + name);
    }
    public static void main(String[] args) {
        Student student = new Student("John Doe");
        student.show();
    }
}
// Output: Name: null
```

```
class Student {
    String name;

    Student(String name) {
        this.name = name;
    }

    void show() {
        System.out.println("Name: " + name);
    }
    public static void main(String[] args) {
        Student student = new Student("John Doe");
        student.show();
    }
}
// Output: Name: John Doe
```

Return the Current Object from a Function

- The '**this**' pointer is often used in **method chaining**

```
class Counter {
    int count = 0;
public:
    Counter& increment() {
        count++;
        return *this;
    }

    void show() {
        cout << "Count: " << count << endl;
    }
};

int main() {
    Counter c;
    Counter d = c.increment();
    Counter e = c.increment();
    e.show();

    Counter().increment().increment().show();
    return 0;
}
```

```
class Counter {
    int count = 0;
    Counter increment() {
        count++;
        return this; // Return current object
    }
    void show() {
        System.out.println("Count: " + count);
    }
}

public class Main {
    public static void main(String[] args)
    {
        Counter counter1 = new Counter();
        Counter counter2 = counter1.increment();
        Counter counter3 = counter2.increment();
        counter3.show();

        new Counter().increment().increment().show();
    }
}
```

Summary

- The '**this**' is a **pointer(C++)/reference(Java) to the current object**.
- It's used inside non-**static** member functions.
- Helps in **disambiguation** and advanced techniques like **chaining**.